Multi-View 2D Image-Based 3D Modeling of Objects: Mid-Semester Progress Report

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***Abstract*—This report presents a complete implementation of 3D object reconstruction from multiple 2D silhouette images using the visual hull algorithm. The methodology involves silhouette extraction, voxel grid construction, visual hull carving, and mesh generation using marching cubes. The system was validated on synthetic objects (cube and sphere) before being applied to a complex diamond object. Experiments were conducted with different rotation angles (0.9°, 1.8°, and 3.6°) to analyze the impact of angular sampling density on reconstruction quality. Results demonstrate that finer angular sampling (0.9°) produces superior reconstruction quality with better detail preservation and fewer artifacts compared to coarser sampling. The visual hull size is determined proportionally to the image dimensions, ensuring correct scaling of the reconstructed 3D model.**

***Index Terms*—3D reconstruction, visual hull, silhouette-based modeling, marching cubes, multi-view geometry, computer vision**

I. Introduction

Three-dimensional reconstruction from multiple 2D images is a fundamental problem in computer vision with applications in robotics, virtual reality, medical imaging, and digital preservation. Taking into consideration the various techniques available silhouette based reconstruction using the visual hull approach is the most efficient technique since it only requires binary object-background segmentation (it requires less processing power) and is less sensitive to lighting changes [2].

Based purely on silhouette information the visual hull is the maximal 3D shape that can be constructed [1]. It is the intersection of all silhouette cones. that is the 3D volumes obtained by back projecting each 2D silhouette into 3D space. The hull of the visible volume is the maximal convex geometry.

This documentation describes the specific end-to-end process of 3D reconstruction from volumetric data (multi-view) silhouette(s) including: (1) silhouette extraction and preprocessing, (2) voxel-based visual hull carving, (3) mesh generation via marching cubes; (4) mesh simplification. The bounding geometry of the shape (cube, sphere, diamond) is obtained performing several rotation scans (0.9°, 1.8° and 3.6°) to observe the effect of sampling on reconstruction quality.

II. Methodology

A. System Overview

The reconstruction pipeline has four main stages: silhouette extraction, voxel grid creation, visual hull carving, and mesh generation. The input consists of a series of images from various points of view around the object, and it is usually a set-up employing a turntable with known rotation steps.

B. Silhouette Extraction and Preprocessing

1) Region of Interest Selection: The first image is interactively selected on a Region of Interest (ROI) to reduce the computational complexity and enhance the quality of silhouettes using OpenCV's `cv2.selectROI` [3]. This ROI is then applied to all follow-up images, ensuring smooth processing of cropping and targeting the object area.

2) Grayscale Conversion and Thresholding: All images are converted to grayscale and binarised using automatic threshold detection (Otsu's method) [3]. It is used to automatically determine the optimal threshold value by analysing the image histogram and finding the threshold that minimises intra-class variance. This adaptive approach ensures optimal silhouette extraction across different lighting conditions and image characteristics without requiring manual threshold selection. The binarisation process converts all images to a binary silhouette, where pixels with an intensity below the threshold are marked as the object (value 1) and others as the background (value 0).

3) Hole Filling: Binary silhouettes often contain small holes due to noise or internal object features. These are filled using morphological operations, specifically `scipy.ndimage.binary_fill_holes` [4]. Thus, this ensures solid silhouettes for accurate visual hull construction.

C. Voxel Grid Construction

The 3D space is discretised into a uniform voxel grid. A normalised coordinate system is created where the object space spans from -1 to +1 along each axis. The voxel grid is created by sampling the 3D space at regular intervals. Each voxel represents a small cubic volume in 3D space. The resolution (typically 196 or 256 voxels per dimension) balances reconstruction detail with computational efficiency (higher resolution provides finer detail but requires more memory and processing time).

Algorithm Overview:

1) Define a normalized 3D space from -1 to 1 along X, Y, Z axes.
2) Sample the space at regular intervals to form a 3D grid.
3) Each voxel represents a small cubic volume in 3D space.

D. Visual Hull Carving

The core of the reconstruction process is visual hull carving [1], as it determines which voxels belong to the object. For each silhouette image captured at a specific rotation angle, the following steps are performed:

1) Viewpoint Transformation: For each image captured at rotation angle $\theta_i = i \cdot \Delta\theta$ (where $\Delta\theta$ is the angular increment), a rotation matrix is constructed to transform 3D coordinates. This assumes rotation around the Y-axis, which is typical for turntable setups.

2) Voxel Projection: Each voxel in the 3D grid is rotated according to the current viewpoint and then projected onto the 2D image plane. The projected coordinates are computed by mapping the normalised 3D coordinates to image pixel coordinates using the image width and height. This mapping ensures that the visual hull size is proportional to the image dimensions.

3) Silhouette Intersection: A voxel stays in the model only if it appears inside the object's silhouette in every image. As we check each view one by one, any voxel that doesn't match is removed. By the end, we are left with the visual hull—the largest 3D shape that encompasses all the silhouettes.

4) Visual Hull Size Determination: The size and shape of the visual hull are determined by the intersection of silhouette cones from all viewpoints. The visual hull size is proportional to the image dimensions, as the projection mapping scales voxel coordinates to image space based on the image width and height. This proportionality is what ensures that the reconstructed 3D model maintains correct aspect ratios and scales appropriately with the input image resolution. Thus, larger images result in a larger visual hull, while smaller images produce a proportionally smaller reconstruction. Other factors that determine the visual hull size include: (1) the number of viewpoints, (2) the image dimensions, and (3) the angular distribution of viewpoints. Using finer angular sampling provides more constraints, which leads to a tighter visual hull that will be more approximate to the actual object.

E. Mesh Generation

The voxel-based visual hull is converted to a polygonal mesh using the marching cubes algorithm [4]. This algorithm extracts an iso-surface from the voxel data by creating triangular faces at the boundaries between occupied and unoccupied voxels. The resulting mesh consists of vertices, faces, and normals. At last, these are exported in OBJ format for visualisation and further processing.

F. Mesh Simplification

The initial mesh from marching cubes often contains excessive polygons. Mesh simplification using quadric decimation reduces the polygon count while preserving shape characteristics [5]. Additionally, the target complexity is matched to a reference mesh to ensure manageable file sizes without significant quality loss.

III. Results

A. Validation on Synthetic Objects

Before applying the system to complex objects, validation was performed on simple geometric shapes to verify algorithm correctness. Synthetic silhouettes were generated for a cube and sphere, representing basic geometries. The reconstruction pipeline successfully recovered their 3D geometry with high accuracy, confirming that:

- The visual hull algorithm correctly handles different object shapes
- Voxel projection and silhouette intersection work as expected
- Mesh generation produces valid 3D models

These validation tests provided confidence to proceed with more complex objects.

Figure 1 shows an example of silhouette extraction from an original image.

(a) Original Image (b) Extracted Silhouette

Fig. 1: Example of silhouette extraction from original image.

B. Diamond Reconstruction

The primary test object was a diamond, a complex shape with both convex facets and concave indentations. Initial reconstruction attempts revealed significant issues where parts of the diamond were incorrectly carved away during the visual hull process, resulting in incomplete geometry with missing regions. This problem manifested as:

- Loss of fine geometric features, particularly at edges and corners
- Incomplete reconstruction of concave regions
- Surface artifacts and irregularities
- Overall reduction in geometric fidelity

These issues were systematically addressed through several algorithmic improvements:

Improved ROI Selection: More precise and consistent region selection was implemented to ensure the entire object is captured across all views. This prevents boundary voxels from being incorrectly eliminated.

Enhanced Thresholding: Adaptive threshold detection using Otsu's method [3] was implemented for better silhouette extraction, accounting for varying lighting conditions across images.

Refined Voxel Projection: Sub-pixel precision with bilinear interpolation was introduced in voxel-to-image mapping, replacing nearest-neighbor sampling. This provides smoother transitions and more accurate silhouette intersection.

Optimized Voxel Resolution: Resolution was increased from 196^3 to 256^3 voxels, providing 30% more spatial detail and enabling finer geometric feature preservation.

Silhouette Quality Enhancement: Morphological operations including noise removal, gap closing, and edge smoothing were applied to improve silhouette quality before visual hull carving.

These improvements collectively resolved the cutting issues and enabled successful reconstruction of the complete diamond geometry.

C. Rotation Angle Analysis

Comprehensive experiments were conducted with three different rotation angles: 0.9°, 1.8°, and 3.6° per image. The results demonstrate a clear relationship between angular sampling density and reconstruction quality:

1) 0.9° Rotation: With 400 images covering 360°, this configuration provides the finest angular sampling. Results show:

- Highest geometric fidelity with fine details preserved
- Smooth surfaces with minimal artifacts
- Accurate representation of concave regions
- Best overall visual quality

2) 1.8° Rotation: With 200 images, this represents moderate sampling density. Results show:

- Good overall shape reconstruction
- Some loss of fine detail compared to 0.9°
- Minor artifacts in complex regions
- Acceptable quality for most applications

3) 3.6° Rotation: With 100 images, this represents coarse sampling. Results show:

- Basic shape is recognizable but lacks detail
- Noticeable artifacts and surface irregularities
- Loss of fine geometric features
- Reduced accuracy in concave regions

Figure 2 shows the reconstructed 3D models for different rotation angles, demonstrating the quality improvement with finer angular sampling.

Table I summarizes the quantitative comparison of reconstruction quality across different rotation angles.

The quantitative comparison reveals that reconstruction quality degrades as angular sampling becomes coarser. The visual hull size increases with coarser sampling because fewer silhouette constraints allow more voxels to satisfy all conditions, resulting in a less accurate approximation. The 0.9° configuration provides the best balance between quality and computational cost for high-fidelity reconstruction.

IV. Discussion

A. Visual Hull Properties and Size Determination

The visual hull is made from silhouettes, with every geometry silhouette represented [1]. Knowing how the size dimensions of the concave geometry outlines and how to manage the volume of these outlines is crucial from a quality viewpoint.

Size Determination Mechanism: The visual hull is defined as the intersection of all silhouette cones from distinct perspectives [1]. Each silhouette cone is the set of all 3D points that project into the corresponding silhouette. The visual hull is defined as voxels that are inside every silhouette of all views. This intersection operation removes voxels that are outside any silhouette.

Proportionality to Image Size: The dimensions of the visual hull are proportional to the size of images. When the coordinates of the voxels are allotted to images, they also consider the width and height of the images. As a result, the visual hull dimensions are proportionately scaled to the size of the input images. Thus the visual hull gets larger as the images are larger and conversely, smaller images would result in a smaller visual hull. In the 3D reconstruction of the object, this proportionality is crucial in preserving the right scale and aspect ratios of the object.

Impact of Angular Sampling: The number and distribution of viewpoints directly affects visual hull size:

- **Fine Sampling (0.9°)**: With 400 viewpoints, each silhouette cone provides a tight constraint. The intersection of many tightly-spaced cones results in a small, accurate visual hull that closely approximates the actual object.
- **Moderate Sampling (1.8°)**: With 200 viewpoints, fewer constraints allow the visual hull to be larger, including more regions that should be carved away.
- **Coarse Sampling (3.6°)**: With only 100 viewpoints, the visual hull becomes significantly larger. Large gaps between viewpoints mean many voxels satisfy all silhouette constraints even though they should be eliminated, resulting in a less accurate approximation.

Finer sampling provides more constraints, resulting in a tighter visual hull that better approximates the actual object. The visual hull volume decreases as the sampling density increases. 0.9° is the gradation where the most accurate and smallest visual hull is observed, followed by 1.8°, then 3.6° hull visualizations respectively.

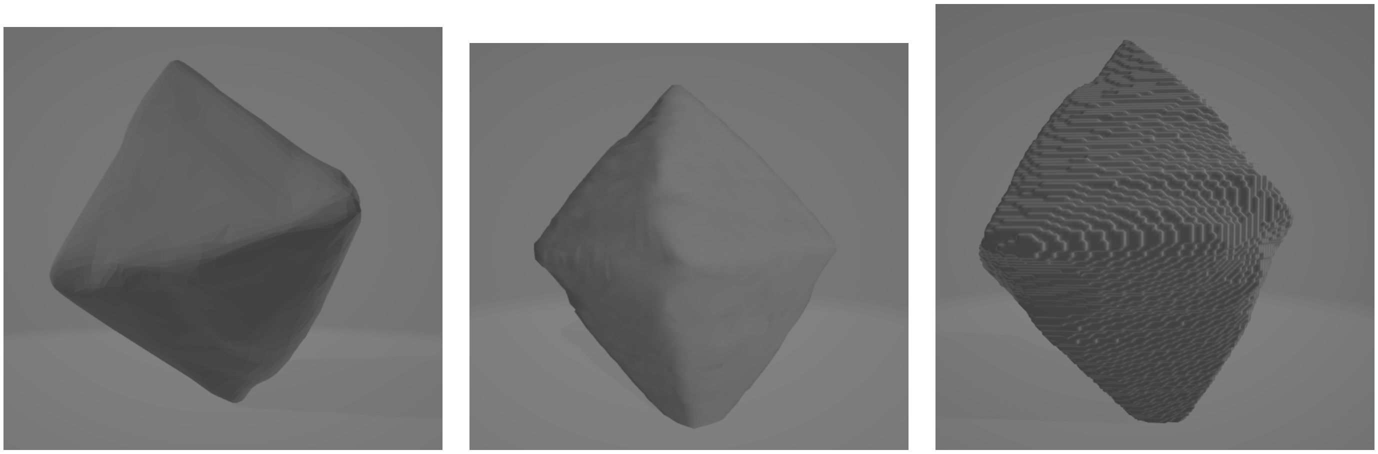
Object Geometry Influence: The way the object is shaped almost always affects the quality of reconstruction. Some concave areas need viewpoints that look into the concavity to be captured correctly. The diamonds have complex geometery with some convex facets and concave indentations, so these outliers need to be captured more often. Visual hull size is proportional to image dimensions, meaning that geometric features are more likely to be accurate with more images.

Silhouette Quality: Accurate silhouettes are crucial. Any errors in binarization or ROI selection propagate through the reconstruction, causing artifacts or missing geometry. The visual hull can only be as accurate as the input silhouettes.

B. Computational Considerations

The computational complexity depends on several factors:

- Voxel resolution: Higher resolution requires more memory and processing time



(a) 0.9° (400 images) (b) 1.8° (200 images) (c) 3.6° (100 images)

Fig. 2: Reconstructed 3D diamond models for different rotation angles.

TABLE I: Comparison of Reconstruction Quality for Different Rotation Angles

Rotation	Images	Quality	Detail	Artifacts
0.9°	400	Excellent	High	Minimal
1.8°	200	Good	Moderate	Minor
3.6°	100	Fair	Low	Noticeable

- Number of images: More images provide better quality but increase computation time
- Mesh complexity: Higher resolution voxel grids produce more complex meshes

Higher resolution and more images improve quality but increase computation time. The 0.9° configuration with 400 images and 256^3 voxel resolution provides the best quality but requires the highest computation.

C. Limitations and Future Work

The visual hull approach has inherent limitations:

- Cannot recover concavities not visible in any silhouette
- Requires complete 360° coverage for best results
- Sensitive to silhouette extraction errors
- Produces outer approximation, not exact geometry

The future improvements that can be included could be: (1) adding photometric to capture surface details, (2) using multi-resolution voxel grids which is more efficient, (3) implementing real-time processing via GPU acceleration, and (4) streamlining automatic parameter optimization.

V. Conclusion

This report presents a complete implementation of 3D reconstruction from multi-view silhouettes using the visual hull algorithm [1]. The system successfully reconstructs complex objects like diamonds, with quality directly correlated to angular sampling density. Experiments demonstrate that 0.9° rotation provides superior results compared to 1.8° and 3.6°, with better detail preservation and fewer artifacts. The visual hull approach offers a robust, computationally efficient method for 3D reconstruction that requires only binary silhouettes [2], making it suitable for various applications where precise geometry is needed but texture information is unavailable or unreliable.

The relationship between angular sampling and reconstruction quality highlights the importance of adequate viewpoint coverage. For applications requiring high fidelity, fine angular sampling (0.9° or better) is recommended, while coarser sampling (1.8°-3.6°) may suffice for applications where basic shape information is sufficient. The visual hull size, being proportional to image dimensions, ensures that reconstructed models maintain correct scale and aspect ratios relative to the input images.

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